Bài Giảng

ROBOTICS VÀ ỨNG DỤNG

(ROBOTICS & APPLICATIONS)

TS. Trần Quang Huy

Transformation

Let u be the point vector representing the point to be transformed,

h be the displacement vector represented by the matrix H, called the transformation matrix,

and v be the vector representing the point after transformation:

We have: $v = H \cdot u$

Translation Transformation

A translation is a type of geometric transformation that shifts every point of an object by the same distance in a given direction.

In matrix form, for a point **u**, the transformation is represented as:

$$v = H \cdot u$$

where:

- **u** is the original point vector (usually in homogeneous coordinates),
- **H** is the translation matrix,
- **v** is the transformed point vector after translation.

Translation Transformation

$$\vec{h} = a\vec{i} + b\vec{j} + c\vec{k}$$

$$H = \text{Trans}(a,b,c) = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$u = [x \;\; y \;\; z \;\; w]^T$$

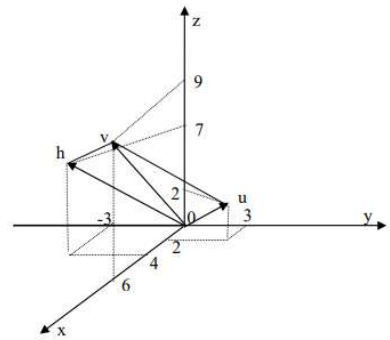
$$v = H.u = \begin{bmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} x+aw \\ y+bw \\ z+cw \\ w \end{bmatrix} = \begin{bmatrix} x/w+a \\ y/w+b \\ z/w+c \\ 1 \end{bmatrix}$$

Translation Transformation

$$\vec{u} = 2\vec{i} + 3\vec{j} + 2\vec{k}$$
$$\vec{h} = 4\vec{i} - 3\vec{j} + 7\vec{k}$$

$$v = Hu = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+4 \\ 3-3 \\ 2+7 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 9 \\ 1 \end{bmatrix}$$

$$v = \text{Trans}(a,b,c)\, u$$



Rotation Transformation

A rotation is a geometric transformation that turns a point or an object around a fixed center (usually the origin) by a specified angle in a specified direction.

In matrix form, for a point **u**, the rotation is represented as:

$$v = H \cdot u$$

where:

- **u** is the original point vector (in homogeneous coordinates if needed),
- **H** is the rotation matrix,
- **v** is the resulting point vector after rotation.

Rotation Transformation

$$Rot(x, \theta^o) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Rot(y, \theta^o) = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation Transformation

$$Rot(z, \theta^o) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation Transformation

Example: Given a point **U** represented by the vector

$\tilde{u}$ = **7i + 3j + 2k**,

rotated around the **z-axis** by an angle **θ = 90°** (Figure 2.5). We have:

$$v = Rot(z, 90^o)u = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 7 \\ 2 \\ 1 \end{bmatrix}$$

If the transformed point is then rotated around the **y-axis** by an angle of **90°**, we have:

$$w = Rot(y, 90^o)v = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 7 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \\ 3 \\ 1 \end{bmatrix}$$

$$w = Rot(y, 90^o).\ Rot(z, 90^o)\ .\ u = \begin{bmatrix} 2 \\ 7 \\ 3 \\ 1 \end{bmatrix}$$

HUYTRAN

ROBOTICS & APPLICATIONS

9

Changing the order of rotation.

$$v' = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -7 \\ 1 \end{bmatrix} = Rot(y, 90^o).u$$

Then, rotating the newly transformed point around the **z-axis** by an angle of **90°**, we obtain:

$$w' = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -7 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ -7 \\ 1 \end{bmatrix} = Rot(z, 90^o).Rot(y, 90^0)u$$

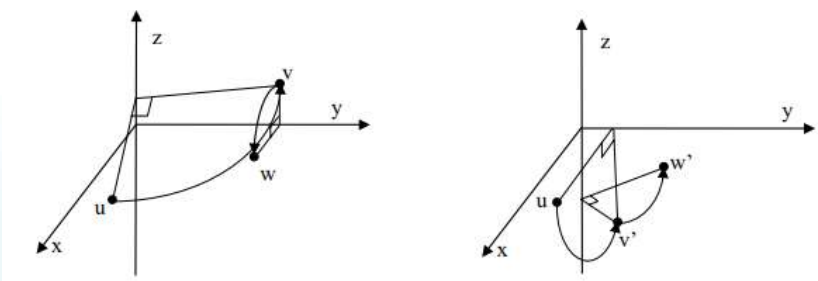
HUYTRAN

ROBOTICS & APPLICATIONS

10

Changing the order of rotation.

$$Rot(y, 90^o).Rot(z,90^0)u \neq Rot(z,90^0).Rot(y, 90^o)u$$



General Rotation

In the previous section, we studied basic rotations around the coordinate axes **x**, **y**, and **z** of the standard coordinate system **O(x, y, z)**.

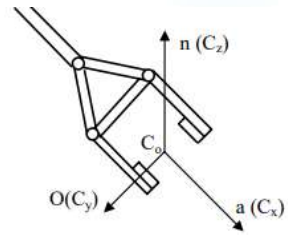
In this section, we examine the rotation around an arbitrary vector **k** by an angle **θ**. The only constraint is that the vector **k** must coincide with the origin of a predefined coordinate system.

Let us consider a coordinate system **C**, attached to the end-effector (hand) of a robot. The coordinate system **C** is represented by:

$$C = \begin{array}{c} \begin{array}{cccc} C_x & C_y & C_z & C_o \end{array} \\ \begin{bmatrix} n_x & O_x & a_z & 0 \\ n_y & O_y & a_y & 0 \\ n_z & O_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

General Rotation

$$C = \begin{array}{c} \begin{array}{cccc} C_x & C_y & C_z & C_o \end{array} \\ \begin{bmatrix} n_x & O_x & a_z & 0 \\ n_y & O_y & a_y & 0 \\ n_z & O_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$



a: the vector pointing in the **approach** direction toward the object;

O: the vector indicating the direction in which the fingers close when grasping the object (**Occupation**);

n: the vector **normal** to the plane formed by (**O**, **a**) — (**Normal** vector).

General Rotation

$$Rot(k,\theta) = \begin{bmatrix} n_x & O_x & a_x & 0 \\ n_y & O_y & a_y & 0 \\ n_z & O_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_x & n_y & n_z & 0 \\ O_x & O_y & O_z & 0 \\ a_x & a_y & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

General Rotation

Nhân 3 ma trận nầy với nhau ta được :

$$
\mathrm{Rot}(k,\theta) = \begin{bmatrix}
n_x n_x\cos\theta - n_x O_x\sin\theta + n_x O_x\sin\theta + O_x O_x\cos\theta + a_x a_x & n_x n_y\cos\theta - n_x O_y\sin\theta + n_y O_x\sin\theta + O_x O_y\cos\theta + a_x a_y & n_x n_z\cos\theta - n_x O_z\sin\theta + n_z O_x\sin\theta + O_x O_z\cos\theta + a_x a_z & 0 \\
n_x n_y\cos\theta - n_y O_x\sin\theta + n_x O_y\sin\theta + O_x O_y\cos\theta + a_y a_x & n_y n_y\cos\theta - n_y O_y\sin\theta + n_y O_y\sin\theta + O_y O_y\cos\theta + a_y a_y & n_y n_z\cos\theta - n_y O_z\sin\theta + n_z O_y\sin\theta + O_y O_z\cos\theta + a_y a_z & 0 \\
n_x n_z\cos\theta - n_z O_x\sin\theta + n_x O_z\sin\theta + O_x O_z\cos\theta + a_z a_x & n_z n_y\cos\theta - n_z O_y\sin\theta + n_y O_z\sin\theta + O_z O_y\cos\theta + a_z a_y & n_z n_z\cos\theta - n_z O_z\sin\theta + n_z O_z\sin\theta + O_z O_z\cos\theta + a_z a_z & 0 \\
0 & 0 & 0 & 1
\end{bmatrix}
$$

HUYTRAN

ROBOTICS & APPLICATIONS

15

General Rotation

To simplify the matrix representation, we consider the following relationships:

- The **dot product** of any row or column of matrix **C** with any other row or column is zero, because the vectors are **orthogonal**.
- The **dot product** of any row or column of **C** with itself is **1**, because each vector is a **unit vector**.
- The unit vector **z** is the **cross product** of vectors **x** and **y**, that is: $\vec{a} = \vec{n} \times \vec{O}$

$$
\begin{aligned}
a_x &= n_y O_z - n_z O_y \\
a_y &= n_x O_z - n_z O_x \\
a_x &= n_x O_y - n_y O_x
\end{aligned}
$$

When vector **k** coincides with one of the unit vectors of the selected frame **C**, we have:

$$k_z = a_x\,;\quad k_y = a_y\,;\quad k_z = a_z$$

$$\mathrm{Vers}\theta = 1 - \cos\theta \quad (\mathrm{Versin}\ \theta).$$

HUYTRAN

ROBOTICS & APPLICATIONS

16

General Rotation

$$\mathrm{Rot}(k,\theta) = \begin{bmatrix} k_x k_x \mathrm{vers}\theta + \cos\theta & k_y k_x \mathrm{vers}\theta - k_z \sin\theta & k_z k_x \mathrm{vers}\theta + k_y \sin\theta & 0 \\ k_x k_y \mathrm{vers}\theta + k_z \sin\theta & k_y k_y \mathrm{vers}\theta + \cos\theta & k_z k_y \mathrm{vers}\theta - k_x \sin\theta & 0 \\ k_x k_z \mathrm{vers}\theta + k_y \sin\theta & k_y k_z \mathrm{vers}\theta + k_z \sin\theta & k_z k_z \mathrm{vers}\theta + \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverse Problem: Determining the Rotation Axis and Equivalent Rotation Angle

Previously, we studied **forward problems**, where the rotation axis and rotation angle are specified in advance, and we observe the resulting transformation.

In contrast, in the **inverse problem**, suppose we are given the result of a certain transformation. Our task is to determine the corresponding **rotation axis k** and **rotation angle θ**.

Assume the result of the transformation is a **homogeneous transformation matrix**:

$$R = \begin{bmatrix} n_x & O_x & a_x & 0 \\ n_y & O_y & a_y & 0 \\ n_z & O_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

We need to determine the rotation axis **k** and the rotation angle **θ**.

$$\begin{bmatrix} n_x & O_x & a_x & 0 \\ n_y & O_y & a_y & 0 \\ n_z & O_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} k_x k_x \mathrm{vers}\theta + \cos\theta & k_y k_x \mathrm{vers}\theta - k_z \sin\theta & k_z k_x \mathrm{vers}\theta + k_y \sin\theta & 0 \\ k_x k_y \mathrm{vers}\theta + k_z \sin\theta & k_y k_y \mathrm{vers}\theta + \cos\theta & k_z k_y \mathrm{vers}\theta - k_x \sin\theta & 0 \\ k_x k_z \mathrm{vers}\theta + k_y \sin\theta & k_y k_z \mathrm{vers}\theta + k_z \sin\theta & k_z k_z \mathrm{vers}\theta + \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverse Problem: Determining the Rotation Axis and Equivalent Rotation Angle

Step 1: Determine the rotation angle θ.

- By adding the diagonals of both matrices on each side, we get:

$$
\begin{aligned}
n_x + O_y + a_z + 1 &= k_x^2\,\mathrm{vers}\theta + \cos\theta + k_y^2\,\mathrm{vers}\theta + \cos\theta + k_z^2\,\mathrm{vers}\theta + \cos\theta + 1 \\
&= (1 - \cos s\theta)(k_x^2 + k_y^2 + k_z^2) + 3\cos\theta + 1 \\
&= 1 - \cos\theta + 3\cos\theta + 1 \\
&= 2(1 + \cos\theta) \\
\Rightarrow \quad \cos\theta &= (n_x + O_y + a_z - 1)/2
\end{aligned}
$$

- Next, we compute the difference of corresponding elements of the two matrices. For example:

$$
\left.
\begin{aligned}
O_z - a_y &= 2k_x\sin\theta \\
a_x - n_z &= 2k_y\sin\theta \\
n_y - O_x &= 2k_z\sin\theta
\end{aligned}
\right\}
$$

- Squaring both sides of the above equations and summing them, we get:

$$(O_z - a_y)^2 + (a_x - n_z)^2 + (n_y - O_x)^2 = 4\sin^2\theta$$

HUYTRAN

ROBOTICS & APPLICATIONS

19

Inverse Problem: Determining the Rotation Axis and Equivalent Rotation Angle

$$(O_z - a_y)^2 + (a_x - n_z)^2 + (n_y - O_x)^2 = 4\sin^2\theta$$

$$\Rightarrow \quad \sin\theta = \pm\frac{1}{2}\sqrt{(O_z - a_y)^2 + (a_x - n_z)^2 + (n_y - O_x)^2}$$

$0 \le \theta \le 180^0$:

$$\mathrm{tg}\theta = \frac{\sqrt{(O_z - a_y)^2 + (a_x - n_z)^2 + (n_y - O_x)^2}}{(n_x + O_y + a_z - 1)}$$

$$k_x = \frac{O_z - a_y}{2\sin\theta}; \quad k_y = \frac{a_x - n_z}{2\sin\theta}; \quad k_x = \frac{n_y - O_z}{2\sin\theta}$$

HUYTRAN

ROBOTICS & APPLICATIONS

20

Inverse Problem: Determining the Rotation Axis and Equivalent Rotation Angle

Example:

Given:

$$R = \mathrm{Rot}[y, 90^\circ] \cdot \mathrm{Rot}[z, 90^\circ]$$

We are to determine the rotation axis **k** and angle **θ** such that:

$$R = \mathrm{Rot}[k, \theta]$$

The resulting transformation matrix is:

$$R = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverse Problem: Determining the Rotation Axis and Equivalent Rotation Angle

$$\sin\theta = \frac{1}{2}\sqrt{(O_z - a_y)^2 + (a_x - n_z)^2 + (n_y - O_x)^2}$$

$$= \frac{1}{2}\sqrt{(1 - 0)^2 + (1 - 0)^2 + (1 - 0)^2} = \frac{\sqrt{3}}{2}$$

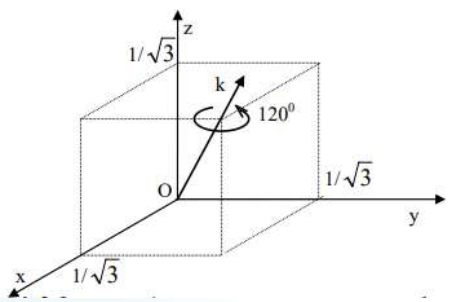
$$\Rightarrow \quad \mathrm{tg}\theta = -\sqrt{3} \quad \text{và} \quad \theta = 120^0$$

$$k_x = k_y = k_z = +\sqrt{\frac{0 + 1/2}{1 + 1/2}} = \frac{1}{\sqrt{3}}$$

Inverse Problem: Determining the Rotation Axis and Equivalent Rotation Angle

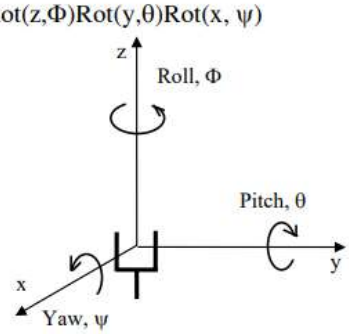
$$R = Rot(y,90^0).Rot(z,90^0) = Rot(k, 120^0);$$

$$\vec{k} = \frac{1}{\sqrt{3}}\vec{i} + \frac{1}{\sqrt{3}}\vec{j} + \frac{1}{\sqrt{3}}\vec{k}$$



Roll – Pitch – Yaw Rotation

$$RPY(\Phi,\theta,\psi)=Rot(z,\Phi)Rot(y,\theta)Rot(x,\psi)$$



Roll – Pitch – Yaw Rotation

$$RPY(\Phi,\theta,\psi)=Rot(z,\Phi)Rot(y,\theta)Rot(x,\psi)$$

$$RPY(\Phi,\theta,\psi)=Rot(z,\Phi)\begin{bmatrix}\cos\theta & 0 & \sin\theta & 0\\ 0 & 1 & 0 & 0\\ -\sin\theta & 0 & \cos\theta & 0\\ 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & \cos\psi & -\sin\psi & 0\\ 0 & \sin\psi & \cos\psi & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}\cos\Phi & -\sin\Phi & 0 & 0\\ \sin\Phi & \cos\Phi & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}\cos\theta & \sin\theta\sin\psi & \sin\theta\cos\psi & 0\\ 0 & \cos\psi & -\sin\psi & 0\\ -\sin\theta & \cos\theta\sin\psi & \cos\theta\cos\psi & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}\cos\Phi\cos\theta & \cos\Phi\sin\theta\sin\psi - \sin\Phi\cos\psi & \cos\Phi\sin\theta\cos\psi + \sin\Phi\sin\psi & 0\\ \sin\Phi\cos\theta & \sin\Phi\sin\theta\sin\psi + \cos\Phi\cos\psi & \sin\Phi\sin\theta\cos\psi - \cos\Phi\sin\psi & 0\\ -\sin\theta & \cos\theta\sin\psi & \cos\theta\cos\psi & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

Coordinate Transformation

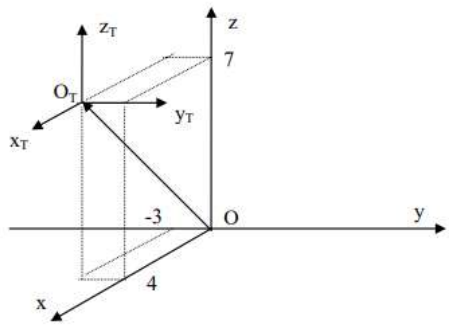
Suppose we need to translate the origin of the Cartesian coordinate system $O(0,0,0)$ by a displacement vector

$$\vec{h} = 4\vec{i} - 3\vec{j} + 7\vec{k}$$

$$O_T = \begin{bmatrix}1 & 0 & 0 & 4\\ 0 & 1 & 0 & -3\\ 0 & 0 & 1 & 7\\ 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}0\\ 0\\ 0\\ 1\end{bmatrix} = \begin{bmatrix}4\\ -3\\ 7\\ 1\end{bmatrix}$$

Coordinate Transformation

This means that the original origin at $O(0,0,0)$ has been transformed to a new origin with coordinates $(4,-3,7)$ relative to the original coordinate system.



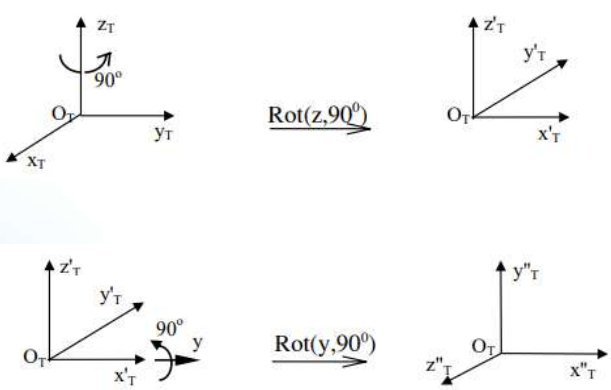
HUYTRAN

ROBOTICS & APPLICATIONS

27

Coordinate Transformation

$\text{Rot}(y,90^o)\text{Rot}(z,90^o).O_T$



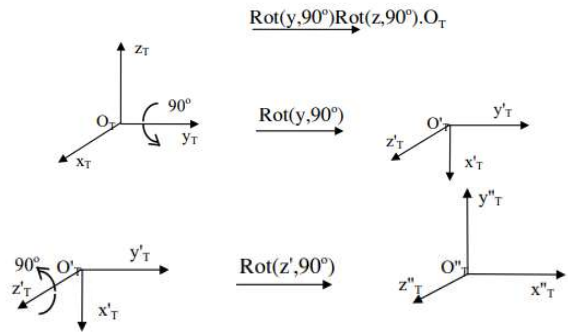
HUYTRAN

ROBOTICS & APPLICATIONS

28

Coordinate Transformation

In the above example, we chose the base coordinate system as the reference frame, and the order of transformations was applied from **right to left**.

If the transformations are instead applied in the **opposite order**, from **left to right**, then the selected reference frame becomes the **intermediate coordinate systems**.

Let us reconsider the example above: